Introduction to Electronic Design Automation

Homework 2

(Problems 1-5)

Student: 吳亞倫
ID: B13901011
Department: 電機工程學系

1 Cofactor

1. (a)

The equalities does not hold for u and v is a literal for the same variable. For example, let $f(x) = x$, $u = x$, and $v = \neg x$. Then, $f_u = 1$ and $(f_u)_v = 1$, but $f_v = 0$ and $(f_v)_u = 0$.

1. (b)

Let v be a literal of variable x. If $v = x$, then

$$(\neg f)_v = (\neg f)|_{x=1} = \neg(f|_{x=1}) = \neg(f_v)$$

If $v = \neg x$, then

$$(\neg f)_v = (\neg f)|_{x=0} = \neg(f|_{x=0}) = \neg(f_v)$$

Hence, $(\neg f)_v = \neg(f_v)$ for any literal v.

1. (c)

Let v be a literal of variable x. By Shannon expansion,

$$f = xf_x + \neg xf_{\neg x}$$

and

$$g = xg_x + \neg xg_{\neg x}.$$

We first prove the case $v = x$. Using Shannon expansion on the function $f \oplus g$, we have

$$f \oplus g = x(f \oplus g)_x + \neg x(f \oplus g)_{\neg x}.$$

On the other hand, substituting the Shannon expansions of f and g into $f \oplus g$, we get

$$f \oplus g = (xf_x + \neg xf_{\neg x}) \oplus (xg_x + \neg xg_{\neg x}).$$

When $x = 1$, this becomes

$$(f \oplus g)_v = (f \oplus g)|_{x=1} = f_x \oplus g_x$$

Next, when $v = \neg x$, setting $x = 0$ in the same expansion gives

$$(f \oplus g)_v = (f \oplus g)|_{x=0} = f_{\neg x} \oplus g_{\neg x} = f_v \oplus g_v.$$

Therefore, in both cases,

$$(f \oplus g)_v = f_v \oplus g_v.$$

Hence $(f \oplus g)_v = (f_v) \oplus (g_v)$ is true.

2 Quantified Boolean Formula

By De Morgan's law, we have

$$\neg(\exists x f(x)) = \forall x(\neg f(x))$$
$$\neg(\forall x f(x)) = \exists x(\neg f(x))$$

2. (a)

For $\Rightarrow$, we have

$$\neg(\forall x, \exists y. f(x, y, z)) \equiv \exists x. \neg(\exists y. f(x, y, z)) \equiv \exists x. \forall y. \neg f(x, y, z).$$

For $\Leftarrow$, we have

$$\exists x. \forall y. \neg f(x, y, z) \equiv \neg(\forall x. \neg(\forall y. \neg f(x, y, z))) \equiv \neg(\forall x. \exists y. f(x, y, z)).$$

Therefore, both directions hold, and hence the equivalence is true.

2. (b)

For $(\Rightarrow)$, consider the following counterexample:

$$f(x, y, z) = (x \wedge y) \vee (x \wedge \neg y).$$

Then $\exists x. \forall y. f(x, y, z)$ is true, since by choosing $x = 1$, we have

$$f(1, y, z) = (1 \wedge y) \vee (1 \wedge \neg y) = y \vee \neg y = 1$$

for all y.

However, $\forall x. \exists y. f(x, y, z)$ is false, since when $x = 0$, we have

$$f(0, y, z) = (0 \wedge y) \vee (0 \wedge \neg y) = 0$$

for every y. Hence, there is no value of y that makes f true.

For $(\Leftarrow)$, consider the following counterexample:

$$f(x, y, z) = (x \wedge y) \vee (\neg x \wedge \neg y).$$

Then $\forall x. \exists y. f(x, y, z)$ is true, since:

- when $x = 1$, choosing $y = 1$ makes f true;
- when $x = 0$, choosing $y = 0$ makes f true.

However, $\exists x. \forall y. f(x, y, z)$ is false, since:

- if $x = 1$, then $f(x, y, z) = y$, which is not true for all y;
- if $x = 0$, then $f(x, y, z) = \neg y$, which is not true for all y.

Hence, neither direction holds in general.

2. (c)

For $(\Rightarrow)$, assume $\exists x. \forall y. f(x, y, z)$ is true. Then there exists some fixed x such that $f(x, y, z)$ is true for all y. Let this x be s. Then, for every y, we have $f(s, y, z)$ is true. Hence, for every y, we can choose that same $x = s$, so $\forall y. \exists x. f(x, y, z)$ is true.

For $(\Leftarrow)$, consider the following counterexample:

$$f(x, y, z) = (x \wedge y) \vee (\neg x \wedge \neg y).$$

Then $\forall y. \exists x. f(x, y, z)$ is true, since:

- when $y = 1$, choose $x = 1$;

- when $y = 0$, choose $x = 0$.

However, $\exists x.\forall y.f(x, y, z)$ is false, since:
- if $x = 1$, then $f(x, y, z) = y$, which is not true for all y;
- if $x = 0$, then $f(x, y, z) = \neg y$, which is not true for all y.

Hence, only ($\Rightarrow$) holds in general.

2. (d)

For ($\Rightarrow$), we use the following implication steps:

1. $\forall x.(f(x, y) \wedge g(x, z))$ (premise)
2. $f(u, y) \wedge g(u, z)$ (UI, 1, with $x = u$ is arbitrary)
3. $f(u, y)$ (simplification, 2)
4. $\forall x.f(x, y)$ (UG, 3)
5. $g(u, z)$ (simplification, 2)
6. $\forall x.g(x, z)$ (UG, 5)
7. $\forall x.f(x, y) \wedge \forall x.g(x, z)$ (conjunction, 4 and 6)

Hence, $\forall x.(f(x, y) \wedge g(x, z))$ implies $\forall x.f(x, y) \wedge \forall x.g(x, z)$.

For ($\Leftarrow$), we use the following implication steps:
1. $\forall x.f(x, y) \wedge \forall x.g(x, z)$ (premise)
2. $\forall x.f(x, y)$ (simplification, 1)
3. $f(u, y)$ (UI, 2, with $x = u$ is arbitrary)
4. $\forall x.g(x, z)$ (simplification, 1)
5. $g(u, z)$ (UI, 4, with $x = u$ is arbitrary)
6. $f(u, y) \wedge g(u, z)$ (conjunction, 3 and 5)
7. $\forall x.(f(x, y) \wedge g(x, z))$ (UG, 6)

Hence, $\forall x.f(x, y) \wedge \forall x.g(x, z)$ implies $\forall x.(f(x, y) \wedge g(x, z))$. Thus, the equivalence is true.

2. (e)

For ($\Rightarrow$), consider the following counterexample:

$$f(x, y) = x \vee y \quad \text{and} \quad g(x, z) = \neg x \vee z$$

Then $\forall x.(f(x, y) \vee g(x, z))$ is true, since

$$f(x, y) \vee g(x, z) = (x \vee y) \vee (\neg x \vee z) = x \vee \neg x \vee y \vee z = 1 \vee y \vee z = 1$$

for every x. However, $\forall x.f(x, y) \vee \forall x.g(x, z)$ is false, since consider $x = 0$, we have $f(x, y) = y$ and $g(x, z) = z$, which are not true for all y and z.

For ($\Leftarrow$):

Assume

$$\forall x.f(x, y) \vee \forall x.g(x, z).$$

Case 1:

$$\forall x.f(x, y)$$

Then for arbitrary u:

$$f(u, y)$$

Hence

$$f(u, y) \vee g(u, z)$$

Thus

$$\forall x.(f(x, y) \vee g(x, z)).$$

Case 2:

$$\forall x.g(x, z)$$

Similarly:

$$\forall x.(f(x, y) \vee g(x, z)).$$

Therefore,

$$\forall x.f(x, y) \vee \forall x.g(x, z) \to \forall x.(f(x, y) \vee g(x, z)).$$

Hence, only ($\Leftarrow$) holds in general.

3 Boolean Function Representation

Consider the majority function $f(x_1, x_2, x_3, x_4, x_5) = (x_1 + x_2 + x_3 + x_4 + x_5 \geq 3)$.

3. (a)

The K-map of f:

- **Case 1**: $x_5 = 1$.

$x_1x_2 \backslash x_3x_4$	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	1	1	1
10	0	1	1	1

- **Case 2**: $x_5 = 0$.

$x_1x_2 \backslash x_3x_4$	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

Hence, the minimal sum-of-products form of f is

$$\begin{aligned} f = {} & x_1x_2x_3 + x_1x_2x_4 + x_1x_2x_5 \\ & +x_1x_3x_4 + x_1x_3x_5 + x_1x_4x_5 \\ & +x_2x_3x_4 + x_2x_3x_5 + x_2x_4x_5 \\ & +x_3x_4x_5. \end{aligned}$$

3. (b)

By turnning the minimal sum-of-products form into a product-of-sums form, we have

$$\begin{aligned} f = {} & (x_1 + x_2 + x_3)(x_1 + x_2 + x_4)(x_1 + x_2 + x_5) \\ & (x_1 + x_3 + x_4)(x_1 + x_3 + x_5)(x_1 + x_4 + x_5) \\ & (x_2 + x_3 + x_4)(x_2 + x_3 + x_5)(x_2 + x_4 + x_5) \\ & (x_3 + x_4 + x_5). \end{aligned}$$

which is the minimal product-of-sums form of f.

3. (c)

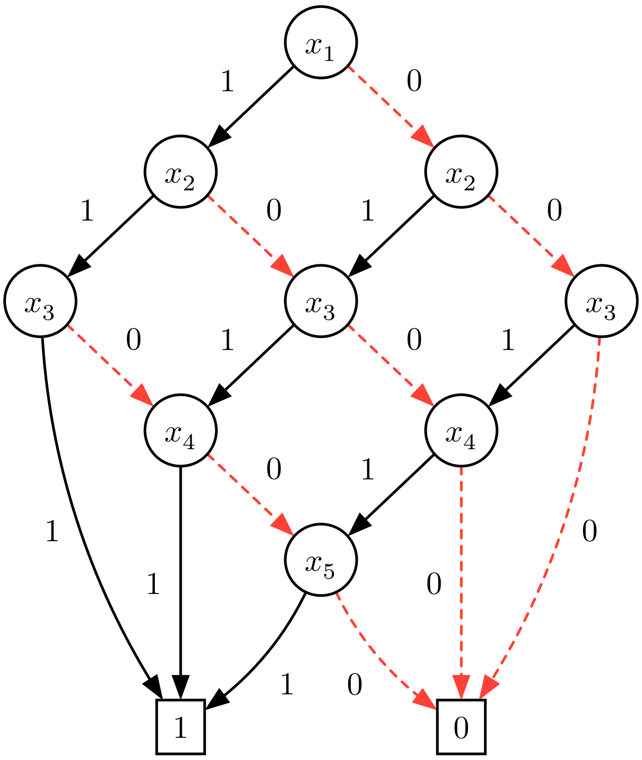


Figure 1: The ROBDD of f.

3. (d)

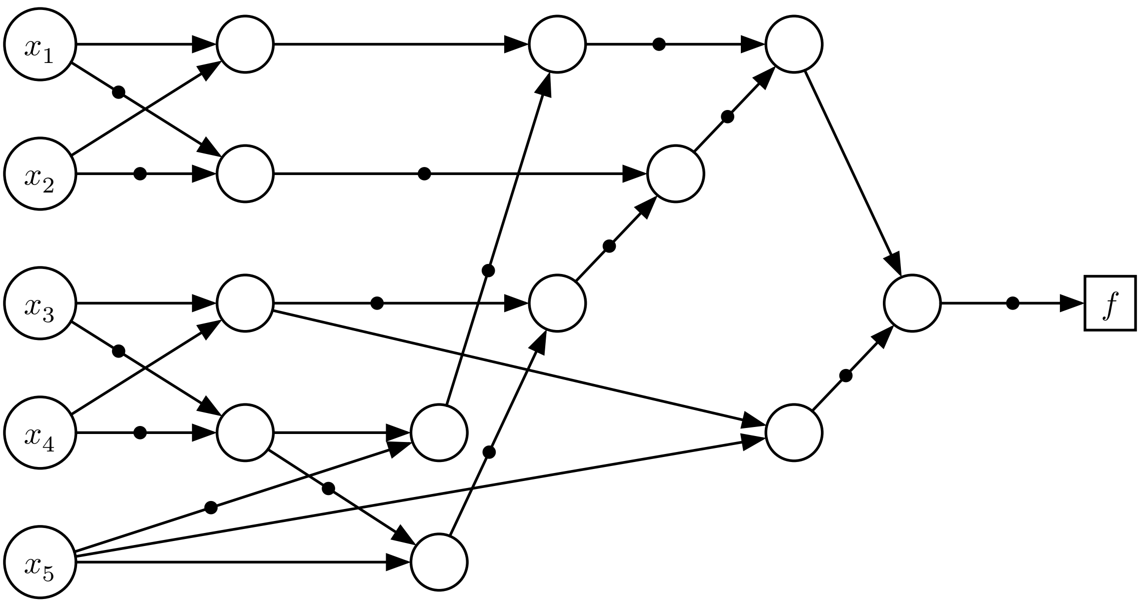


Figure 2: The AIG of f. Use total 12 nodes.

3. (e)

We can express f as:

$$f = M(x_1,\ M(x_2, x_3, x_5),\ M(x_4, x_5, M(x_1, x_2, x_3)))$$

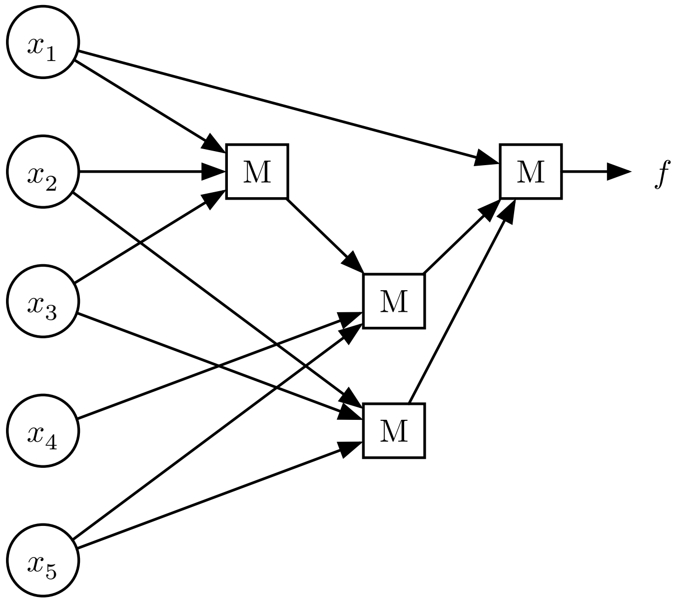


Figure 3: The circuit of f using 3-input majority gates. Use total 4 gates.

4 Application of Quantified Boolean Formula

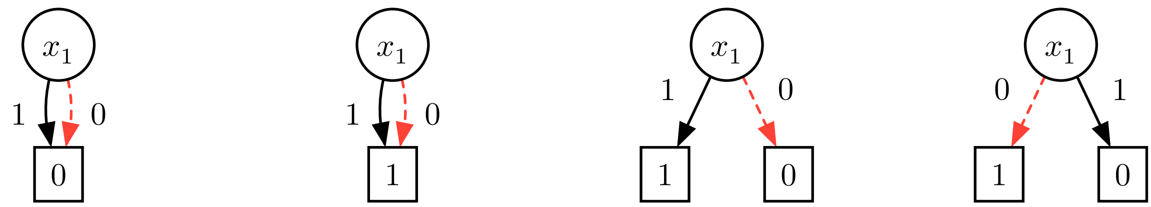
5 Binary Decision Diagram

5. (a)

Let $f(x_1, x_2, ..., x_n)$ be a any n-input Boolean function. WLOG, we can assume the variable ordering is $x_1 < x_2 < ... < x_n$. We want to show that the ROBDD of f is unique. We prove this by induction on n.

Base case: $n = 1$. In this case, f can only be one of the following four functions:
1. $f(x_1) = 0$;
2. $f(x_1) = 1$;
3. $f(x_1) = x_1$;
4. $f(x_1) = \neg x_1$.

The ROBDD of each of these functions is unique as shown below:



$f(x_1) = 0$ $f(x_1) = 1$ $f(x_1) = x_1$ $f(x_1) = \neg x_1$

Induction step: Assume the statement holds for any Boolean function for $n = 1, 2, ..., k - 1$. We want to show that the statement also holds for $n = k$. Let $f(x_1, x_2, ..., x_k)$ be an n-input Boolean function. By Shannon expansion, we have

$$f = x_1 f_{x_1} + \neg x_1 f_{\neg x_1}.$$

By the definition of ROBDD, the root node is decided by the first variable x_1. The high child of the root node is the ROBDD of f_{x_1}, and the low child of the root node is the ROBDD of $f_{\neg x_1}$.

By the induction hypothesis,f_{x_1} and $f_{\neg x_1}$ are $k-1$ input Boolean functions, hence the ROBDDs of f_{x_1} and $f_{\neg x_1}$ are unique. Therefore, the ROBDD of f is also unique, since the root node is unique and the high and low children are unique. This completes the induction step.

Thus, by induction, the ROBDD of any n-input Boolean function is unique.

5. (b)

$$f = \bigwedge_{i=1}^{n} (x_i \vee y_i)$$

Case 1: $O(n)$ nodes

Consider the order of variables to be $x_1 < y_1 < x_2 < y_2 < \ldots < x_n < y_n$. We can construct an ROBDD of f having $O(n)$ nodes as shown below:

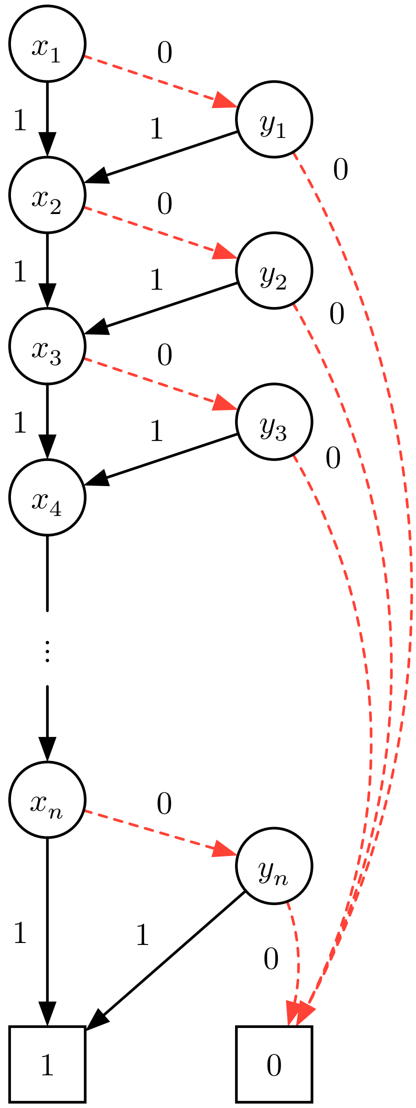


Figure 4: An ROBDD of f with $O(n)$ nodes.

Case 2: $O(2^n)$ nodes

Consider the variable order $x_1 < x_2 < \ldots < x_n < y_1 < y_2 < \ldots < y_n$. We show that the ROBDD of $f = \bigwedge_{i=1}^{n}(x_i \vee y_i)$ has $O(2^n)$ nodes.

First, consider the part of the ROBDD before any y-variable is tested. Since the variables $x_1, \ldots, x_n$ are tested first, this part forms a binary decision tree of height n. Hence it contains $2^n - 1$ nodes.

Next, after all $x_1, \ldots, x_n$ are assigned, the remaining function is determined by the set of indices i such that $x_i = 0$. For each assignment $(x_1, \ldots, x_n) = (a_1, \ldots, a_n) \in \{0, 1\}^n$, the residual function is $\bigwedge_{i:a_i=0} y_i$. Since there are only 2^n possible subsets of $\{1, \ldots, n\}$, there are at most 2^n distinct residual functions. Because an ROBDD merges identical subgraphs, the part below the x_n-level contains at most 2^n distinct subgraphs.

Therefore, the total number of nodes is at most $(2^n - 1) + 2^n = 2^{n+1} - 1 = O(2^n)$. Hence the ROBDD has $O(2^n)$ nodes.

An example of such an ROBDD with $n = 3$ is shown below:

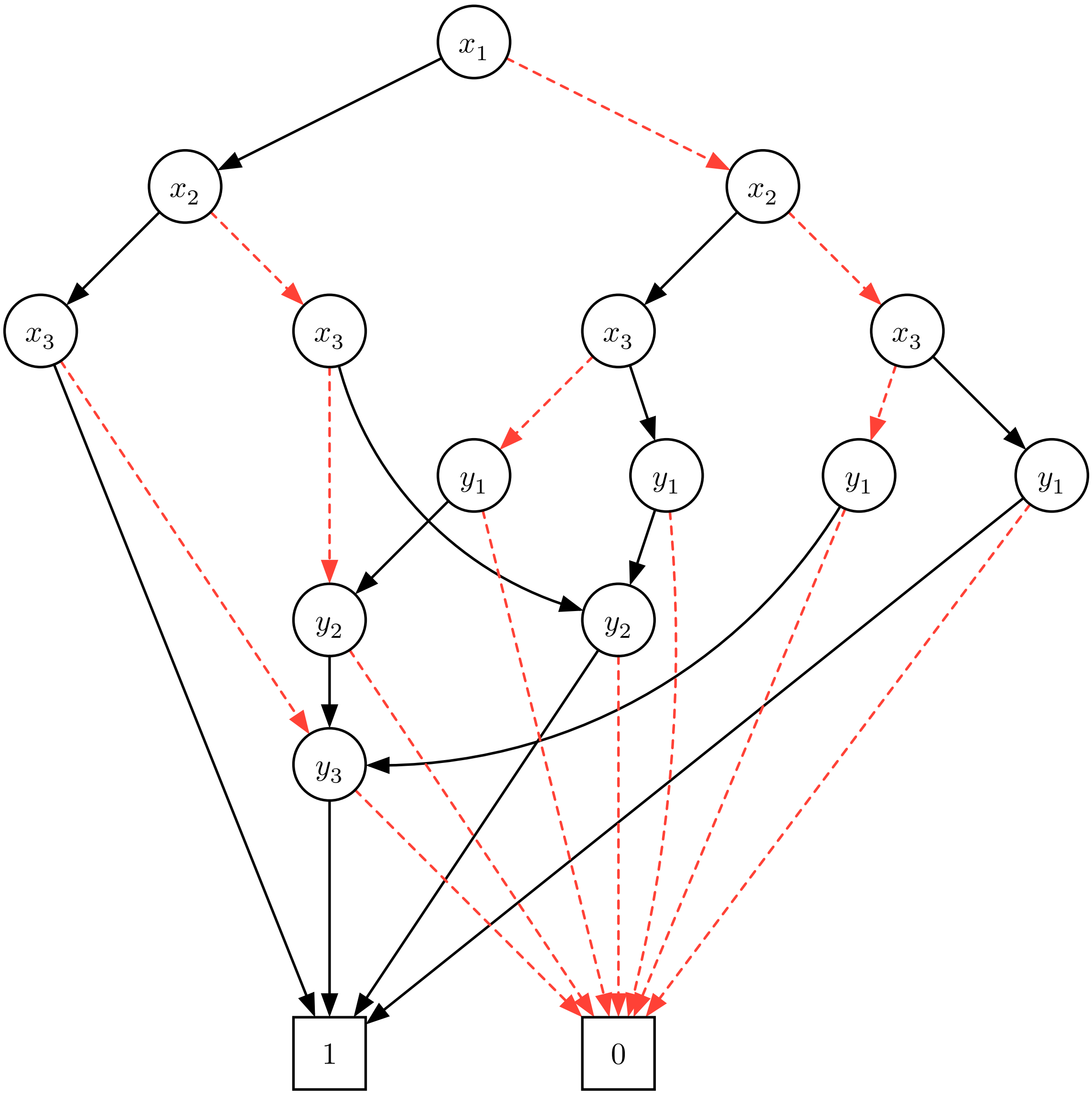


Figure 5: A ROBDD of f with $O(2^n)$ nodes. In this example, $n = 3$ and the ROBDD has 15 nodes.